LSRII Startup, Quality Control and Shut Down Procedure
UC Davis Flow Cytometry Shared Resource
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https://ccresources.ucdmc.ucdavis.edu

Startup Procedure:

1. **Check sheath and waste levels before powering on the cytometer.**

   Saline supply and waste collection are managed on the LSRII by the FACS Fluid Supply System pump unit located on the floor under the LSRII. This pump unit will pull saline into the smaller pressure reservoir and push saline through a 0.22um filter to the machine and flow cell. Waste fluid, consisting of saline plus analyzed sample, will be pushed down into the waste box. The waste box contains 1L of bleach, as sample is added, it comes into contact with the bleach and is decontaminated.

   Before turning on the power, check the fluid levels

   Make sure sheath cube is at least ½ full, and that the waste is no more than ½ full.
   If more saline is needed, get a new saline box from the storage area in room 3427 located adjacent to the core by the sorting suite. Install the new box of saline at the FACS
   - Add 1000mL (1L) of bleach to bottom of waste tank
   - Reinstall waste tank at cytometer

2. **Turn on LSRII main power**
3. **Turn on the FACS fluid Supply pump below LSRII, silence the alarm**
   - Turn on instrument a minimum of 30 minutes prior to performing instrument QC.
   - Verify that instrument is in “standby” mode.

4. **Login to host HP computer: Windows login Administrator, pw: leave blank**
5. **Prepare the fluidics system by purging any air bubbles from the saline supply line circuit**
   1. Purge bubbles from sheath line filter
      i) Get a waste bucket
      ii) Hold sheath filter upright, inspect closely for air bubbles
      iii) Tilt filter so that bubble is near exit tubing port, open the roller clamp, keep tubing over bucket to collect ejected sheath fluid, make sure bubble exits to waste bucket, retighten roller clamp
   2. Purge bubbles from sheath tubing between filter and LSRII entry port:
      i) Place a waste bucket under roller clamp, release fluid and any bubbles into waste tank, retighten roller clamp, empty bucket at sink, and hang on peg drying rack
   3. Purge bubbles from the flow cell:
      i) With sample support arm to the side (you will hear the Droplet Containment Management pump begin to cycle), press the prime button on the front control panel of the LSRII. The prime procedure evacuates the flow cell laser intercept chamber and sends all the sheath fluid in the flow cell to waste, then reestablishes sheath flow up through the cuvette. This process purges any

stubborn air bubbles from the interrogation chamber. Keeping the sample tube support arm to the side, repeat the purge cycle two more times, for a total of 3 prime cycles.

4. Run water to chase any remaining air from the fluid lines:
   i) Replace a tube of DI water on the SIP, engage sample support arm, place fluidics control panel in RUN, with HIGH sample air pressure, wash flow cell and push any air bubbles in the waste line to the waste tank by running plain DI H20 for 5 minutes while you start up the software and cytometer.

6. Launch BD FACSDiva software 6.1.3 from desktop icon
7. Login to DIVA user account
8. Perform Cytometer Setup and Tracking (CST) automated QC routine

***** (See note below on re-using previously prepared CST beads)**

1) Prepare a fresh tube of CST beads – be sure to use the current bead lot – the bead lot must be selected in the CST software module to allow the automated quality control check to proceed.
2) Obtain a clean FACStube: write bead lot and date on tube
1) Invert or vortex CST bead vial to re-suspend settled beads
2) Add 300ul of sheath fluid or PBS to the tube, add 1 drop of CST beads to the tube, vortex gently before running the performance check
3) Set fluidics mode to low, at the midpoint setting – see below under shutdown procedure for an explanation of the sample pressure modes and sample injection rates.
4) It is important to achieve a event rate of about 500 events/second when running CST beads or the performance check will fail – so don't dilute the beads too much or the event rate will be too low.
5) ***Diluted preparations of CST beads can be re-used within 1-3 days of preparation – so please put the diluted CST bead sample back in the CST bead box for the next user.
6) When performance check is completed, be sure to view the performance report and check the laser delay timing and target voltages and CV's for the 3 sentinel detectors:

- Violet laser = Cascade Blue
- Blue laser = FITC
- Red laser = APC

***Leftover diluted CST beads can be saved and re-used within 2-3 days of preparation if they are capped, and stored in the CST bead box, protected from light in the 4C walk-in refrigerator next to the flow core. Please re-use these diluted beads. Make fresh beads only when needed.

Shut Down Procedure:

It is important to wash the fluidics system after each use to remove residual sample and fluorochromes from the system, this may be immediately necessary if sticky, clumpy samples (such as samples from digested, disaggregated tissue or large cultured cells) are run.

1. Obtain 3 FACS tubes (also, it is ok to re-use previously prepared washing tubes, often these are in the tube rack near the LSRII) and <u>fill ¾ full with the following</u>:

- Tube #1: 10% bleach in H20
- Tube #2: BD FACSrinse solution, if available, or dilute Contrad (10% solution in H20)
- Tube #3: DI water.

2. Turn instrument to run, high flow rate, and turn fine adjustment knob all the way to the right to achieve the fastest flow rate possible (see below).

   Midpoint at each flow rate setting: 5 clockwise (right) turns of the fine adjustment knob

   - Low: 12ul/min, range is 6-24ul/min
   - Med: 35ul/min, range is 17.5-70ul/min
   - High: 60ul/min, range is 30-120ul/min

3. Install the 10% bleach tube on the SIP with the support arm to the side, so that the vacuum pump is engaged and bleach is aspirated quickly up the outer sleeve of the SIP. Run bleach until only 1mL of fluid remains in tube – this takes about 1 minute.
4. Move sample support arm directly under tube so that the remaining 10% bleach can be run up the center of the SIP and into the flow cell. Run for 5 minutes.
5. Repeat this procedure with the FACSRinse or Contrad tube, and finally the DI water.
6. Leave water tube on the sample station.
7. Choose "STANDBY" on the instrument's fluidic control panel.
8. Turn off power to the instrument only if no other users are scheduled to use the instrument.
9. Turn off power to the Fluidics Cart under the instrument.

Troubleshooting:

Changing the sheath or emptying waste during an acquisition session:

Turn off the power to the FACS Fluid Supply System (the pump unit on the floor underneath the LSRII).

To add saline: Add saline (use a funnel) or install a new box of saline, save any residual unused saline for Flow Core staff, cap and set aside.

To empty waste: start by removing the lid, unscrew from box and set aside, transfer box to sink, empty waste fluid into sink, refill (use a funnel) with one liter of bleach, replace waste box on floor and reconnect to FACS Flow Supply System.

Clogs – it is possible to clog the silver sample injection tube where it tapers to a fine point inside the flow cell. Typical signs are your sample suddenly stops showing data on the DIVA screen or your FSC and SSC signals suddenly shoot off scale in all directions.

To unclog the LSRII:

Remove your sample from the sample station, set it aside.
Push the sample support arm to the side
Prime twice using the prime button on the fluidics control panel on the front of the LSRII
Prepare a tube of **50%** bleach in DiH20, run on high, with sample pressure set to HIGH
Be patient – clogs can take time to remove
Remove the 50% bleach tube. Prime the flow cell twice.
To rinse out the strong bleach solution, run a tube of plain DiH20 on HIGH for 2-3 minutes.
To determine if the clog was removed, use the CST bead tube (using old dilute beads is fine here).
Place the diluted CST beads on the sample station and hit run, try to visualize the CST beads on the flow plots (adjust FSC to 500V to see beads, they are SMALL!). Using beads, which are small 3um plastic spheres and much smaller than cells, will help you to determine if the clog has been dissolved.

Laser delay problems: typical manifestations – red fluorochromes do not appear to be giving proper signals. Solution – remove your sample. Check the sheath supply tubing for bubbles. Ensure that the FACS Fluid Supply

System is on and that the pressure vessel of sheath is actually working. Prime the flow cell 2-3 times, run a tube of DiH20 on high run for 2-3 minutes and re-run CST.

CST warnings and failures:

Warning: CST will return a warning message (!) if the coefficient of variation (CV) for the 3um brightly dyed bead fluorescence varies by more than 6% for three key detectors. If the CV for the bright CST bead is higher than 6% in the Pacific Blue (405nm laser, 450nm emission), FITC (488nm laser, 530nm emission), and APC (638nm laser, 670nm emission) the CST software will highlight the increased CV in red font and produce a passing result with a "warning". What this means is that the LSRII is having difficulty detecting bright, energetic fluorescence emissions. Several factors could contribute to the imprecision of these bright signals on the LSRII and can be addressed by specific corrective steps:

1. Dirty flow cell – intervention: run a tube of dilute Contrad detergent as a sample on high run for 5 minutes. Follow with water, then re-run CST.
2. Bubbles in sheath line and/or flow cell – intervention: check fluidics path…is there sufficient sheath in supply box? Is the FACS Flow fluid supply system pump unit powered on? Is there saline in the pressurized sheath tank? Bubbles in the supply sheath line? Remove the sample tube from the sample station, place the sample support arm to the side and run the "Prime" cycle 3 times (sample rate control panel on front of LSRII).
3. CST bead dilution degraded due to storage in dilute state for too many days – intervention: make a fresh aliquot of CST beads in 300ul diluent (water, sheath)
4. Lasers not warmed up: The LSRII's lasers need ~30min of warmup time. Inadequate/insufficient warmup time can yield poor CVs. Wait a few more minutes and re-run CST.
5. Laser alignment may be drifting: If the CST reports over several days continually return warnings or failures, please alert the core staff so that a service visit can be scheduled.

CST Failure message (**X**):

CST will produce a failure message if any detector voltage must be either increased or decreased by 50V as measured against the "baseline" detector voltages. Baseline detector sensitivity and recommended operational voltages are determined when the cytometer has undergone a service visit – during the service visit the lasers are typically realigned to improve illumination of the sample in the flow cell. This ideal operational state of the flow cytometer's optical and fluidic systems is captured by the core facility staff through the CST "baseline" routine – core staff use the CST bead mixture to conduct a thorough assessment of detector sensitivity and resolution immediately after